Conclusion

A national-scale superintelligence project would be a massive undertaking. The stakes would be high, the margins of error slim, and the infrastructure costs staggering. It would invite adversaries like China to accelerate their own programs and try to slow down ours. We'd need to move fast and retain control of a system that's as smart or smarter than we are, while under constant threat from peer-level geopolitical adversaries. But this is arguably the trajectory we're on anyway: China is a live player in the race to ASI, they're already spying on all our top labs, and they're already pushing as hard as they can to develop domestic supply chains for everything from AI chips to frontier AI models. It's not clear that the CCP could be much more forward-leaning on ASI than they already are.

The window to kick off a project is also closing. In private conversations, frontier AI researchers tell us they believe we're less than a year away from having AI agents that can automate the *majority* of current software engineering work. We may not be that far from superintelligence already, and data centers take time to build. Existing data centers are nowhere near as secure as they'll need to be, and CCP-proof security can't be retrofitted.

There's no guarantee that superintelligence will come soon. But if it does, and if we plan to prevent the CCP from stealing it, we might need to break ground on a fully secure, gigawatt-scale AI data center within the next few months for a project like this to be relevant once the build is complete. And before we can do that we'll need to determine what good security looks like for superintelligence clusters. This is all doable, but every day we don't move on it limits the options we'll have available to us when things heat up.

And while nobody knows the complete picture yet, we can at least say a few things about what a project like this would have to look like and what its high-level objectives should be.

Objective of the project

A domestic superintelligence project would have a huge energy footprint that we wouldn't be able to conceal from adversaries.[60] This was a universal point of agreement among everyone we spoke to, from the intelligence community to the DOE to domain experts in AI hardware and energy.

So if we build a superintelligence cluster, China will know about it. And because they'll know about the project from the start, we should expect they'll both accelerate their own competing projects and also try to disrupt ours in any way they can. China has already announced a $275B investment in AI infrastructure,[61] within days of the announcement of $100B in committed funding for Project Stargate. If they're serious about this – and anybody who drops a quarter trillion dollars is serious[62] – we can expect that the lead times on generators, transformers, and other China-source components for U.S. data centers will start mysteriously lengthening. This will be a tell that China is quietly diverting components to its own facilities, since after all, **they control the industrial base that is making most of them.** If they're clever about it, they will of course never announce this as a policy — we'll just see annoying delays and chalk it up to competitive bidding for rush orders, ordinary horse-trading, mundane "supply problems", and so on.

China is a serious contender in AI, and they're already beating us in key parts of the AI supply chain. They exclusively manufacture many of the prefabricated components and inputs to AI

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[60] This is true if we put the superintelligence cluster in a single region. But if we tried to decentralize the cluster — by spreading it across multiple data center campuses in different states, for example — intelligence specialists have told us that the communication links between the campuses would be too vulnerable to be viable, even if they involved dedicated point-to-point connections over dark fiber.

[61] The investment was 1T yuan which is equivalent to about $275B when measured in PPP terms, which is the right comparison for a project that's acquiring equipment and talent in-country. While the official USD to CNY conversion rate was about 7.3 CNY per USD, in PPP terms it's estimated closer to 3.64 CNY per USD — so this investment is more than twice as big as it looks.

[62] There are other reasons to believe the CCP is making superintelligence a national priority. Most recently, DeepSeek reportedly required certain employees to hand in their passports to prevent them from travelling abroad, because their work made them, "privy to confidential information that could constitute trade secrets or even state secrets". Headhunters who have tried to poach DeepSeek employees have also reportedly received calls from local governments telling them to stop, and investors looking to make pitches to DeepSeek have been told to reach out to Zhejiang Province's Communist Party committee to be screened first – a particularly noteworthy requirement given that China is otherwise trying to reverse a collapse in foreign direct investment.

data centers in the U.S., and they have far more grid capacity available for advanced AI training and inference. Chinese companies have genuinely world-class engineering teams,[63] and they're making meaningful progress despite American export controls.

This means that a U.S.-led superintelligence project has to be grounded from the beginning in the unforgiving reality of a global AI race between superpowers. And because we and our adversaries will be racing for what could be a decisive strategic advantage, this race could escalate without limit across all warfighting domains. That potentially could include kinetic conflict or even nuclear war, but it's more likely to be a game of sabotaging and disrupting each others' competing AI projects to stop the other side from making too much progress.[64] Of course, this kind of sabotage is just as likely if we continue to leave development of frontier AI completely in the hands of private companies instead of kicking off a national project.

In a race like this, the only way we can succeed is for **the United States to establish and maintain a substantial lead over its adversaries** — particularly China — as we build ever more powerful AI systems on the path to ASI. That lead is critical whether your main concern is the weaponization of advanced AI, the potential for loss of control over a superintelligence, or the risk of unchecked centralization of power.

Let's walk through these one after the other:

- This is obviously true for **weaponization**. In the hands of a nation that's able and willing to deploy it, a controlled superintelligence is an insurmountable strategic advantage. The consequences of arriving second in this race could be dire — and adversaries will be thinking this too.

- If your main concern is that we might **lose control** over superintelligent systems, then maximizing lead time is just as critical. The longer our lead time is, the wider our window will be to develop the technical tools we'll need to reliably control a superintelligence. Although if we get very close to achieving superintelligence before we've developed those tools, pushing forward could create more problems than it solves. This is also a reason to ensure that a national project prioritizes developing the

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[63] For example, when it launched in January 2025, DeepSeek R1 famously matched the performance of OpenAI's o1 – OpenAI's most advanced model then available to users – across a wide range of key benchmarks.

[64] This could lead to a new kind of (potentially unstable) equilibrium between major AI powers. Based on private correspondence with Dan Hendrycks. Aspects of this are discussed in the MAIM framework co-developed by Dan.

narrowest possible tools [redacted] that will provide the United States a lasting advantage over the CCP, unless and until we're confident that we can solve the control problems that come with building broad superintelligence.

- Finally, if you care most about **concentration of power**, then a CCP-controlled superintelligence is the most dangerous possible scenario. A U.S.-led project is much more likely to develop superintelligence in an environment with checks and balances, public oversight, and controls to stop a single person or small group from seizing power. Of course that means it's our responsibility to build these features into the project from the start. But a CCP-controlled superintelligence has no chance of having any such controls.

The CCP is a dictatorship and Xi Jinping is its dictator. That means a CCP-controlled superintelligence will end up being a Xi Jinping-controlled superintelligence, no matter how much the CCP or its surrogates may deny it. And a Xi Jinping-controlled superintelligence is likely also the worst outcome *for the Chinese people* themselves — which includes China's political elite, up to and including the members of their Politburo — given that they might no longer have economic value to Xi Jinping personally.

As superintelligence approaches and this reality begins to dawn on the highest levels of the Chinese political elite, it could lead to internal power struggles – maybe even coup attempts – within the CCP. As high-ranking officials begin to realize that their positions, influence, and even personal safety might be at risk, we should be on the lookout for signs of instability. America may find opportunities to exploit the CCP's internal power dynamics to disrupt their control over domestic AI development.[65]

Strategies

So the top priority of a U.S.-based superintelligence project is clear: we'll have to establish and maintain a substantial lead over the CCP and all other adversaries. There are three things we'd need to do to accomplish this:

1. First, we'll need to actively **delay the CCP's progress** towards superintelligence. This delaying effort will cover many domains and involve some of the most powerful

[65] The more credible our own claim to having legitimate democratic oversight over our AI project can be, the stronger our hand will be in taking advantage of these opportunities.

capabilities that can be deployed by our national security agencies. Because we don't know how difficult it will be to control a superintelligence, we would need to sustain and extend our lead by any means necessary; we can't assume that just accelerating our domestic AI development will give us the lead time we will need. [redacted]

2. Second, we need to **secure our existing frontier AI R&D** against theft by adversaries. Until and unless we do this, **we don't have *any* real lead in the AI race over the CCP**. As we've already seen, our national security agencies aren't free to spy on our domestic companies, so it's more likely that advances in American frontier AI labs translate into CCP military AI capabilities *before* they translate into U.S. national security capabilities. Even if we decide not to start a full scale national superintelligence project, we need to put excellent security in place around our existing AI labs.

3. Third, we would need to **make R&D progress towards controlled superintelligence — but only *after* we have secured it**. Without strong security, faster U.S. AI progress will empower the CCP as much as, or more than, it will empower the United States. Only after we have *established* a lead through strong security, can we hope to *sustain* and *extend* that lead.

A lot of people shrink from proposing options such as aggressively delaying our adversaries' progress, because it brings a risk of escalation, and it does. But whether the United States pursues a national superintelligence project or not, simply continuing along our current path of AI development *already* carries escalation risk. You can't build what you claim is a de facto superweapon without drawing a response from nation state adversaries as you approach your goal. Right now, if an American frontier lab closes in on superintelligence, the odds are that they'll have their model weights stolen and their training infrastructure sabotaged (possibly even without their knowledge) to prevent them from completing their project, leaving a U.S. adversary decisively ahead. In fact, some of the security specialists we spoke to, including former NSA officers currently specialized in frontier AI security, see this as **the most likely default outcome** of our current trajectory.

Slowing down the CCP's progress on AI is already a core pillar of U.S. strategy. And it will only become more important given that the CCP currently has structural advantages over the United

States in power production,[66] data center component production, and several relevant areas of infrastructure. According to the CEO of DeepSeek, the main current limitation to Chinese progress in AI are U.S. export controls on AI computing hardware — but Chinese champions like Huawei, SMIC, and CXMT appear to be making progress onshoring critical supply chain components. The CCP also has a greater ability to concentrate hardware and industrial activity in support of a Chinese government-backed project than the U.S. government does.

The three lines of effort above will reinforce each other even if we don't pursue a full-fledged national superintelligence project. As we develop more powerful models, [redacted] Whatever course we end up taking, improving our intelligence-gathering capabilities with respect to Chinese AI progress should be a top priority. This approach might provide us with a valuable option that currently doesn't exist: the choice not to build superintelligence at all, if it seems unlikely that we'd be able to control it. [redacted] Some people suggest cooperating with China or Russia in this area. Unfortunately, no national security professional we've spoken to takes this idea seriously today. The trust gap is just too great and spans too many areas. To give just two examples, Russia started to systematically violate the Intermediate-Range Nuclear Forces (INF) Treaty in the mid-2000s, to the point where it now has active battalions with medium-range nuclear forces. And China is building up a global nuclear arsenal that will soon rival America's alongside "Beijing Military City", a massive underground continuity of govt complex big enough to house 10 Pentagons. According to several sources, China is unwilling to talk about or even acknowledge their nuclear buildup in diplomatic engagements. These countries need to do a lot more to earn America's trust before there's any chance of cooperating with them on a strategic issue like superintelligence.

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[66] And under current conditions, this will only get worse. There are currently 29 nuclear power plants under construction in China, out of 65 total globally.

Cooperation would have to involve technical solutions that allow us to verify that our adversaries are actually upholding their commitments to us. Tamper-proof on-chip monitoring and remote shut-down capabilities might qualify, but these capabilities are years away. Any arrangement that requires America to trust the CCP or other adversary regimes is a non-starter under current conditions. That said, the project could be used to support the development of trustless treaty enforcement solutions (including flexHEG-like technologies) that would provide options for creative "ASI diplomacy" down the line. While we should be looking for opportunities to use powerful forms of AI to de-escalate geostrategic tensions, we can't lose sight of the fact that the CCP is a highly capable adversary that sees itself as being locked in a struggle with the United States, and which has been more than happy to take advantage of American goodwill and play dirty with diplomacy in the past.

Project ownership

If the U.S. government started a superintelligence project, which part of the government should own it?

We spoke to officials from dozens of departments and agencies about this, and it's become clear to us that if a national-scale project *is* going to be overseen by one department, it will probably have to be the Department of Energy.[67] The DOE has limited experience with the kinds of supercomputers it takes to train modern AI, but they do have experience with supercomputers in general — and they have expertise in infrastructure, security, nonproliferation, along with energy.[68] They also have a basic pipeline that allows them to prototype (via APRA-E) and scale new programs and technologies, and the capacity to sustain larger projects than, e.g. DAPRA.

By and large though, most energy, defense, and national security specialists we talked to — including people within the government itself — didn't think the government would do a good job operating a project like this day to day. Frontier AI is just way too fast-paced and unpredictable, and requires fast, high-conviction deployments of technical talent and capital that the government isn't equipped for. A government department could play a leadership role

[67] With support from DARPA, CYBERCOM, the NSA, and CIA, among others.

[68] As one option, we could fund DOE to sponsor a consortium of FFRDCs that would oversee the project.

in a project like this, but its role should be carefully scoped, first to high-level supervision, and second to close monitoring of the key factors that aren't related to AI capabilities: security, control, and the integrity of the project's chain of command.

Whether or not a government-coordinated national project goes forward, the government definitely needs trusted communication channels directly with technical AI researchers, especially the ones who are working on AI control. We hear repeatedly that some labs are carefully filtering the information they provide to the government about what's really going on and what they intend to do. National security decision-makers need a way to bypass policy teams and lab executives and talk directly to the people who are doing the real work – including whistleblowers.

Wherever it ends up, the project will need to collaborate closely with several other groups. At the very least it will need to be integrated closely with elements of the intelligence community for security, counterintelligence, and [redacted]. It will need organizations with specialized expertise in AI auditing, evaluations and control, and research institutions like universities. And realistically, a project like this will have to be set up as a public-private partnership. That means collaborating closely with the frontier AI labs that have the technical expertise to build superintelligence, if that's possible in the near future.

Project funding

This will be a big project. Who will pay for it? It'd be hard for the government to pay for the whole thing. Hyperscalers are expected to spend $300 billion a year on infrastructure in 2025, with most of that being AI related. That's already almost 20% of federal discretionary spending. If you add the security, control, and governance needs that a project like this will have, we should expect it to cost significantly more than that per year. What's more, a superintelligence project that starts today will have to use resources, companies, and talent that already exists. Those resources have already been funded by investors who expect a return. And private sector incentives have done a great job so far at incentivizing AI capabilities development anyway.[69]

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[69] Congress *could* appropriate an unprecedented amount of funds for a project like this. But in order for that to become a politically realistic scenario, we'd probably need warning shots of AI capabilities so obviously WMD-like that the window of opportunity to kick off a project with reasonable security will have closed.

The best strategy we've heard to solve this problem is to allow investors to receive returns from some of the outputs of a project like this to keep them incentivized to fund increases in AI capabilities.[70] For example, if the project develops a next-generation AI model that it knows can be sold securely, then it can distribute the profits from the model's use back to the original private sector investors.[71] We could also offer preferred tax treatment to AI investors to incentivize their continuing participation if the project needs more funding.

At the same time, the government itself is going to need to directly fund the security, AI control, and oversight functions of the project that investors aren't incentivized to support. Private sector incentives don't get you good enough security — we hear stories time and time again from our contacts about how truly terrible the security at frontier AI labs is. Private sector incentives push companies to cut corners on AI control too, turning "alignment" into a product metric (or even worse, a metric associated with political goals, like tuning the political slant of their models) rather than investigating the growing evidence that we're having trouble controlling the systems we're building. And the kind of oversight we need to prevent a project like this from being corrupted needs to be much better thought out than the strategies we use to govern private companies.

Of course, any major government funding for a project like this will need Congress's buy-in. We'll also need a special funding channel so that project resources are flexible enough to avoid the problems that our budget has every fiscal year.[72]

Security, espionage, and counterintelligence

In our conversations with special forces operators, as well as intel and national security specialists, one thing was emphasized over and over: you can't expect to protect a national

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[70] And to ensure those investments are concentrated in a single project.

[71] For example, you can imagine allocating revenue based on market cap at the time of a potential merger between labs. That brings a real risk of regulatory capture, but by that point, only the largest labs may be able to participate anyway.

[72] "No-year funding" is probably the best way to set this up. If you have no-year funding, that means the funds that you don't spend in one fiscal year are still available in the next year, whereas with ordinary funding, you either use it or lose it. It tends to be hard to get no-year funding because Congress gives up some of its power (over your budget) by giving it to you. But a superintelligence project can't afford to have its funding frozen with Continuing Resolutions every year — it's bad enough already that we do that with our defense appropriations.

superintelligence project with ordinary security alone. The only way to have even a chance of securing a project like this is to combine it with an offensive counterintelligence team that's constantly seeking out external and internal threats — and **actively operating against them**, including with offensive operations.

That means there's no version of a national superintelligence project that isn't integrated with active, offensive operations right from the beginning. It's an uncomfortable fact, but there's no path to success where we build a perfect fortress and develop superintelligence behind its walls — we need to be reaching out and touching our adversaries from day one.[73] Every world class operator we spoke to agreed about this: until and unless you actually use a capability against an adversary in the real world, you don't know if your capability is real or not. Acting is a big risk, but waiting is an even bigger risk. There's no way out but through.

A big part of this project involves actively degrading and disrupting CCP, and possibly other adversary AI capabilities and projects, and that introduces some escalation risk. On the other hand, developing superintelligence, *whether we do it as part of a national scale project or a private project*, is probably enough of a threat to those adversaries to provoke eventual retaliation in and of itself.[74] Besides, if we take a hands-off approach and let a private company develop superintelligence without government support, we ought to expect that the CCP will wait until a U.S. company is close to achieving it, and then steal the company's model and sabotage their training run and physical infrastructure, before finishing the superintelligence project themselves to deploy it against us. When we say that **the United States does not have any real lead in the AI race** until we secure our AI labs, that's what we mean.

On top of active counterintelligence, ordinary security for the project needs to be excellent and set up fast. There are some quick wins to be had here. We should localize data centers for the project at greenfield sites in geographically secure areas, for example in Middle America, and staff them with cleared contractors (or take measures to achieve equivalent levels of security).[75] The more space there is between the data center sites and urban areas, the harder it is for

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[73] This is even more true to the extent that our adversaries are making significant, independent progress on algorithms and hardware design. DeepSeek's V3 and R1 models make it clear that this is happening in China.

[74] What's more, Russian and Chinese operations are engaged in significant, low-level industrial sabotage on U.S. soil already, as many national security professionals have highlighted to us privately.

[75] That doesn't necessarily mean contractors that have actual security clearances: the security clearance process as implemented by the government takes way too long to be practical. You can do a *better job* with private sector background check solutions *if you pick the right providers*, and do it much faster.

adversaries to observe or infiltrate, and the easier it is to secure. A data center for a superintelligence project will also become a magnet for sabotage and even kinetic attacks by adversaries, so the further it is from inhabited areas, the less collateral damage we'll absorb if things go badly. There's also an information security advantage to consider: leaky municipal record-keeping, FOIA requests, and other channels could easily reveal security-critical information.

Even if you build your data center as an exquisite fortress that's 100% impenetrable (which is already impossible), you still haven't protected your model's weights from theft. Among many other options, an adversary can simply take out your exquisite data center through sabotage. And if they do, you're dead in the water and can't make any further progress in the AI race — *unless* you transfer your work to a less secure data center. You're then faced with a dilemma: allow your adversaries to pull ahead, or resume your work using infrastructure that wasn't designed to withstand nation-state theft operations. In other words: if you're in a race to superintelligence, there's no way to protect your model weights from theft without *also* protecting your data center from destructive attacks.[76]

There are a whole host of potential attacks that could accomplish this. For example, as we've already seen, there are straightforward attacks that can take down multi-billion dollar pieces of (relatively secure) infrastructure for six months on a $20,000 budget. It's a fairly cheap fix at a cost of a few hundred thousand dollars. But fixes like those need to be implemented. Nothing's perfect, but we should at least force our adversaries to dig deeper than $20k if they want to brick some of the biggest investments in our country's history.

But some critical fixes are much more expensive. If we're serious about security, we'll need to redesign data centers to resist TEMPEST attacks,[77] which, among other things, involves physically separating servers from the inner and outer walls of the building — in other words, a fully redesigned campus footprint. We'll also need to secure the supply chains for components

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[76] But if you're *not* in an AI race, this is no longer true. For example, if the United States and China are just trying to *prevent* each other from achieving superintelligence rather than pushing hard to achieve it themselves — possibly because they don't believe they themselves understand how to control a superintelligent AI. Then a destructive attack on your data center might temporarily stop you from improving or even using your most advanced AI model, but won't destroy the model itself. (It's relatively easy to at least preserve a model's weights securely.) Thanks to Dan Hendrycks for suggesting this point.

[77] This is where an attacker can read what a computer is doing from its electromagnetic emanations. The actual standards used to defend against TEMPEST attacks are classified and presumably rapidly evolving – another reason why a project like this would require close collaboration between intelligence agencies and data center design teams.

that go into the data centers to the same degree that's currently being done for warfighting platforms like fighter jets. Yet according to data center operators we spoke to, some of those components do not have non-Chinese suppliers. And if the superintelligence cluster needs to be distributed geographically, the fiber links between buildings need to be continuously monitored for physical access and compromise[78] — according to experts we spoke to, [redacted]. There are lots of other measures needed too but these are the most under-discussed.[79]

We're aware of several teams currently researching whether it's possible to retrofit an existing data center to this level of security. Their preliminary conclusion is that it isn't doable. That's a big problem: we need this level of security or we immediately lose the game. So a project like this needs a new data center build, it needs security baked in from the ground up, and it needs to be integrated from the start into an offensive counterintelligence motion that disrupts similar projects by adversaries. As new capabilities come online that can buy America time, they need to be quickly and efficiently folded into offensive activities, while intelligence we collect from adversaries needs to inform the *kinds* of capabilities we seek to develop in real-time.

We need a dedicated function to balance security, AI control and containment, and capabilities progress, informed by classified intelligence and a deep understanding of the risks of leaks and uncontrolled superintelligence. And there's no way to accomplish that other than to put competent operators in charge, who have clear visibility into all the relevant factors.

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[78] Even dark fiber used exclusively for this purpose are apparently vulnerable to these attacks. Encrypted fiber links could help here, but we suspect that either RSA encryption has been broken today, or that national security agencies anticipate that it could be broken imminently. This suspicion is based on recently observed public statements by facets of the U.S. government; we don't have any private information about this particular topic.

[79] Some of the most prominent ones include screening of all devices entering data centers; auditing source code on all cluster devices (especially network cards and AI accelerators); dedicated protected storage for models and sensitive assets with monitorable external connections and flexible network isolation (allowing for offline operation); mandatory encryption of sensitive data at rest and in transit within the cluster; secure identity and integrity attestation for all connected devices; encrypting model weights at all times except when in use and in volatile memory only; defining network boundaries between sub-networks within the facility and between the facility and outside world, with pre-defined network baseline configurations for data flows at the boundaries and flags for anomalous traffic; and continuous monitoring of inter-data center connections (including underground cables). Hardware and compute security should include strong anti-tampering measures, confidential computing at the cluster level, protection against side-channel attacks, and hardware-level protection of monitoring logs. Physical access should be strictly limited to cleared personnel with high-end professional security services, regular sweeps for unauthorized devices, and full isolation of hardware.

All of this can be done, and we've surfaced what we think are workable solutions to many of these problems throughout. But they'll call upon us to do more than business as usual: to take the reins of history, aim high, and execute with extreme competence. Luckily, that's exactly what America does best.